

Git for Science

A substrate for reproducible, compounding scientific knowledge

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Git did not just store files. It stored the *repository*: a structured, content-addressed object with identity and lineage, of which the working files are one view. That object is why merging, provenance, and collaboration became mechanical, and why a planet's worth of code could accumulate in one shared form. Science is still pre-git. We ship the paper, the printout of a result, and throw the object away. This is an attempt to store the object: what a result *is* (a typed physical quantity, its formula, its lineage, and the code that produced it as one representation of it), so that results become reproducible by construction, knowledge compounds instead of evaporating, and an AI agent has a semantic place to act.

The creed

1. **The unit of scientific knowledge is meaning, not text.**
2. **Physics is the invariant. A code is one representation of it.**
3. **A scientific result is a structured object with identity and lineage, not a number in a table.**
4. **A result you cannot re-run is a claim, not yet knowledge.**
5. **Units, gauge, and lineage are part of a result, not footnotes to it.**
6. **Knowledge should compound, not evaporate into scripts.**
7. **Correctness should be structural, checked by the substrate, not re-derived by trust.**
8. **Verification must scale with compute, not with reviewers.**
9. **The native substrate for scientific AI is semantic, not token text.**

The problem: science that does not accumulate

Computational materials science is a Tower of Babel. A single physical quantity, say the lattice thermal conductivity of a crystal, is computed by a dozen codes: kaldo, phono3py, ShengBTE, LAMMPS, VASP, Quantum ESPRESSO. Each has its own input formats, its own unit conventions, its own gauge and basis choices, and its own unspoken assumptions about what was held fixed. The physics is the same. The codes never quite agree, and deciding whether two of them *should* agree is itself expert work.

That work does not accumulate. Reconciling two calculations is artisanal, done by a handful of people who carry the conventions in their heads, and most published computational results are effectively irreproducible because the real workflow lives in tribal knowledge

and one-off scripts rather than in the paper. The paper is the tarball we email. Once it is sent, the object that produced it is gone.

The diagnosis: meaning is implicit

The reason the codes do not agree mechanically is that their meaning is implicit. Scientific knowledge today lives in three forms, and each loses something. Prose, in papers, carries rich meaning but does not run. Code runs, but its meaning is entangled with implementation: the physics is mixed in with array layouts, loop orders, and file formats. Data is numbers stripped of their provenance. None of the three carries meaning that is at once machine-checkable and runnable.

So the physics is never stored directly. It is buried inside whichever artifact happened to carry it, and recovering it, deciding what a given number actually claims, is left to the reader. What cannot be recovered mechanically cannot be checked mechanically, and so it cannot accumulate.